Worksheet No. - 5

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Aim/Overview of the practical:

To gain hands-on experience in creating and using cursors for row-by-row processing in a database, enabling sequential access and manipulation of query results for complex business logic. **(Company Tags: Infosys, Wipro, TCS, Capgemini)**

Objective:

- **Sequential Data Access:** To understand how to fetch rows one by one from a result set using cursor mechanisms.
- **Row-Level Manipulation:** To perform specific operations or calculations on individual records that require conditional procedural logic.
- **Resource Management:** To learn the lifecycle of a cursor: Declaring, Opening, Fetching, and importantly, Closing and Deallocating to manage system memory.
- **Exception Handling:** To handle cursor-related errors and performance considerations during large-scale data iteration.

Input/Apparatus Used:

- **Oracle**
- **MS SSMS**
- **PostgreSQL**

Procedure/Algorithm/Code :

```
select

   a.machine_id,

   ROUND(AVG(b.timestamp - a.timestamp), 3) as processing_time
```

```
from Activity a

join Activity b

on a.machine_id = b.machine_id

and a.process_id = b.process_id

and a.activity_type = 'start'

and b.activity_type = 'end'

group by a.machine_id;
```

Output:

	machine_id integer	processing_time numeric
1	0	0.894
2	1	0.995
3	2	1.456

Learning outcomes (What I have learnt):

- Understand how to use **EJB (Stateless Beans)** for implementing business logic in enterprise applications.
- Learn to apply **Bean Validation annotations** to ensure data integrity before storing in the database.
- Gain knowledge of **JPA Entity mapping** and how objects are persisted in relational databases.
- Understand and implement **One-to-One relationship mapping** using @OneToOne and @JoinColumn.
- Develop the ability to write **SQL queries for data analysis**, including joins, aggregation, and handling database-specific errors.